

Amey Abhay Joshi

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Education	Michigan State University: PhD in mathematics	(ongoing)
	College of Natural Sciences	
	Michigan State University: Masters in Mathematics	Dec'25
	College of Natural Sciences	
	IISER Pune : Master of Science	May '22
	IISER Pune : Bachelor of Science	May '22
	Vivekanand College, Kolhapur: 11th and 12th standard	July '15-June '17
	S.M.Lohia high school, kolhapur: 5th-10th standard	June '09-April '15
Employment	MSU-Teaching Excluded Assistant II	Aug22-May23
	MSU-Teaching Assistant MTH234	Summer 23, Fall 23
	MSU Summer Topology Program Mentor	Summer 23
	MSU-Teaching Assistant MTH 314- Matrix Algebra with Applications	Spring 24
	MSU-instructor of record - MTH314	Summer 24
	Mathematics Instructor- Louis Stokes Alliance for Minority Participation (LSAMP)	
	Summer 24	
	Grader - MTH327H	Fall 24
	TA MTH314	Spring 25
	Lead TA MTH124	Spring 25
	Lead TA- for first year grad students	Summer 25
	iSTEM Math instructor	Summer 25
	Graduate Real Analysis TA	Fall 25
	Lead TA - MTH 124 Survey of calculus	Fall 25

Research Interests Khovanov Homology, Heegard-Floer, Symplectic Topology, Four manifold Theory.

Publications [Abelian branched covers of rational surfaces](#)
with Robert Harris, Doug Park, Mainak Poddar
In Preparation: Spectral Sequences in Lasagna theory

Research projects

Advisor

- Khovanov Homology and quantum error-correction [with Zaiku Group](#)
- Inverse problems in Machine Learning Prof. Mark Iwen
- Skein Lasagna Modules and Spectral Sequences Prof. Matthew Stoffregen
- Kirby color for TQFTs and knot theory based invariants Prof. Matthew Hedden, Prof. Matthew Stoffregen
- Sheaf Theoretic applications of Atiyah Floer Conjectures Prof. Matthew Hedden

	• Four Manifolds and Gauge Theory MS Thesis project	Doug Park
	• Symplectic Classes of representations	Steven Spallone
organizational and Volunteer experience	• Co-organizer : Graduate Student Topology and Geometry Conference 2024 • Mathematics Mentor - Louis Stokes Alliance for Minority Participation Summer 2024 • Organizer- Kirby Color and Skein Lasagna Modules Seminar Fall 2024 • Volunteer- Math night for Elementary School Fall 2024 • Instructor and Project Guide: Computational applications of matrix algebras- Student Projects Summer 2024 • Organizer- Deeper structures in Lasagna Modules of TQFTs Spring 25	

Reading Projects

Advisor

• Kirby color for Khovanov Homology	Matthew Stoffregen, Matthew Hedden
• Floer Homology	Matthew Stoffregen, Matthew Hedden
• Four Manifolds and Gauge Theory	Prof. Doug Park
• h-cobordism theorem semester project SPRING 2021,	Prof. Mainak Poddar
• Characteristic Classes and Milnor's exotic sphere Summer 2020	Prof. Mainak Poddar
• Lie Groupoids and Orbifolds as a part of VSRP 2020	Prof.S.K.Roushon
• Schemes, Complex Manifolds and non-algebraic manifold mini project of Algebraic Geometry course SPRING 2021	Dr. Vivek Mallick

Conferences and Workshops-

- [Workshop- Scottish Talbot on Algebra and Topology](#)
- Links in 3 and 4 dimensional Manifolds, ICERM, May 2025
- International Georgia Topology Conference, Athens, May 2025
- New structures in low dimensional topology, Budapest 2024 (online participation due to technical travel issues)
- 41st Workshop in Geometric Topology 2024
- [Trisectors Workshop](#) 2023
- Introductory Workshop: Floer Homotopy Theory Sep'22
- ["Geometric Topology"](#) conference Dec-2019.
- [IISc-IISER math symposium](#) Sep-2021.
- [Unifying Themes in Geometry](#) Sep 2021.

Talks	• Diagramatic non-vanishing result - STOAT'26 • Inverse problems in Electro Magnetic Theories - April' 25 • Abelian Branched Covers of Rational Surfaces- 41st Geometric Topology workshop • "Khovonov Homology returns" (How to get manifold invariants from knot invariants) at MSU student topology seminar Fall'24 • organizer and main presenter - "kirby color for Khovonov Homology" seminar Fall'24 • Symplectic Geography Problem - MSU student topology seminar Fall'23 • Lagrangian Floer Homology - Fall'23 • Morse-Bott Theory- Fall'23 • Kirby Color in Khovonov Homology - Series of weekly talks Fall'24
Awards and Honors	• TA Award for Excellence in Teaching by MSU math department Spring'26 • Douglas A. Spragg Endowed Fellowship Award Spring'26 • Graduate Research Associate Fellowship Spring'26 • Infosys foundation scholarship 2021-22 • TIFR Visiting Student Research Program fellow Summer 2020 • IISER Mohali Summer Research fellow Summer 2019 • National Graduate Physics Exam 2018-19: national top 1 percent • DST-INSPIRE SHE fellow since 2017 • IIT-JEE Advanced 2017 All India Rank-5056 (equivalent to 99 percentile)
References (Industrial Research)	• Bamborde' Balde' , Founder:Zaiku Group • Prof. Mark Iwen, Department of Mathematics
References (Research)	• Prof. Matthew Hedden, Department of Mathematics, MSU • Dr. Matthew Stroppgen, Department of Mathematics, MSU • Prof.Doug Park, Department of Mathematics, University of Waterloo, Ontario, Canada. email : bdpark(AT)uwaterloo.ca • Prof.Mainak Poddar, Chair, Department of Mathematics, IISER Pune. email : mainak(AT)iiserpune.ac.in • Dr.Steven Spallone, Associate professor, Department of Mathematics IISER Pune. email : sspallone(AT)iiserpune.ac.in
References (Teaching)	• Tsvetanka Sendova, Department of mathematics, MSU. • Jamillah C Gross-Caldwell, specialist and LSAMP coordinator, MSU
Software	• Python • SageMath • L^AT_EX • MATLAB • HTML

Languages

- English : Fluent
- Hindi : Fluent
- Marathi: Mother Tongue
- Sanskrit: High school elective